IGCSE 0607 Practice Questions — E7.3 — Combined Questions & Markscheme

Exact trigonometric values

Total Questions: 9 — **Total Marks:** 52 Marks

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- **E7.3** Exact trigonometric values
  → **Paper 2 (Non-Calculator)** *(9 Qs)*

1 Without using a calculator, answer the following questions.

(a) Write down the exact value of $\cos 30°$.

..... **[1]**

(b) Show that

$$\frac{1-\sin 30°}{\tan^2 60°} = \frac{1}{6}$$

[2]

2 Without using a calculator, write down the exact value of each of the following expressions.
You must show all your working.

(a) $\sin 90^\circ + \cos 0^\circ$

..... **[1]**

(b) $\tan 45^\circ \times \sin 30^\circ$

..... **[1]**

(c) $\dfrac{\tan 60^\circ}{\cos 30^\circ}$

..... **[2]**

3 **(a)** Work out the exact value of

$2\sin 60° \times \tan 30°$

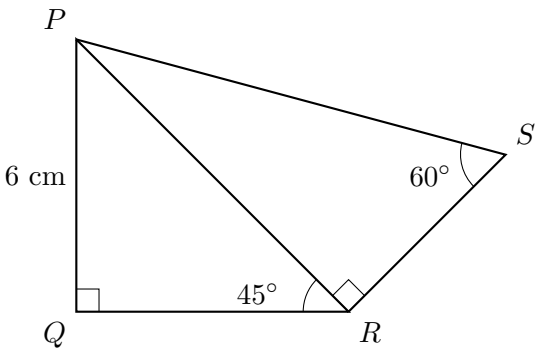
..... **[2]**

(b) Find the value of x, where $0° \leq x \leq 90°$, such that

$\sqrt{2}\cos x = \tan 45°$

$x =$ ..... ° **[2]**

4 The diagram shows two right-angled triangles, PQR and PRS.
$PQ = 6$ cm, angle $PQR = 90°$ and angle $PRQ = 45°$.
Angle $PRS = 90°$ and angle $PSR = 60°$.



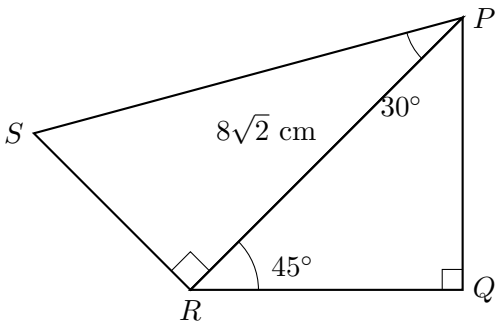
(a) Calculate the exact length of PR.

..... cm **[2]**

(b) Calculate the exact length of PS.
Give your answer in the form $a\sqrt{b}$ where a and b are integers.

..... cm **[3]**

5 The diagram shows two right-angled triangles, PQR and PRS.
Angle $PQR = 90°$ and angle $PRS = 90°$.
Angle $PRQ = 45°$ and angle $SPR = 30°$.
$PR = 8\sqrt{2}$ cm.



NOT TO SCALE

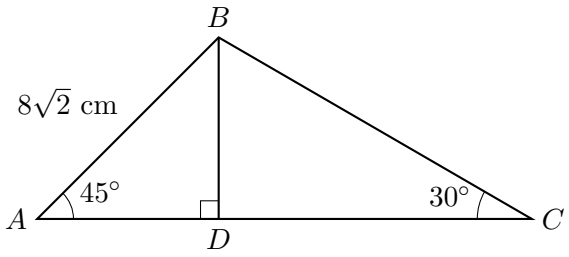
(a) Without using a calculator, show that the exact length of PQ is 8 cm.

[2]

(b) Without using a calculator, work out the exact length of PS.
Give your answer in the form $\frac{a\sqrt{b}}{c}$, where a, b and c are integers.

$PS =$ ..... cm **[3]**

6 The diagram shows a triangle ABC with a point D on the side AC.
BD is perpendicular to AC.
$AB = 8\sqrt{2}$ cm, $\angle BAC = 45°$ and $\angle BCA = 30°$.



Calculate the exact perimeter of triangle ABC.
Give your answer in the form $a + b\sqrt{2} + c\sqrt{3}$ where a, b, and c are integers.

Perimeter = ... cm **[6]**

7 Do not use a calculator for this question.
All working must be clearly shown.

(a) Show that

$$\frac{1-\tan^2 30^\circ}{1+\tan^2 30^\circ} = \cos 60^\circ$$

[3]

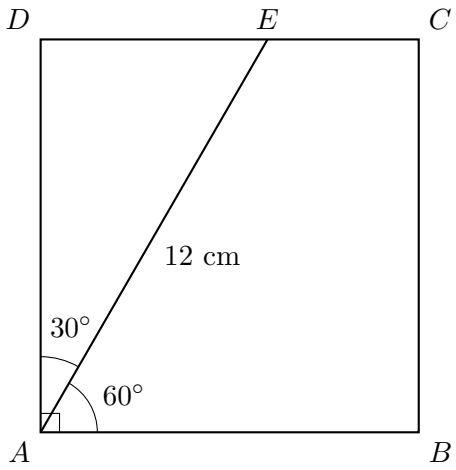
(b) Solve the equation.

$$x \sin 60^\circ = \tan 60^\circ - x \cos 60^\circ$$

Give your answer in the form $a - \sqrt{b}$, where a and b are integers.

$x =$ ..... **[4]**

8 The diagram shows a rectangle $ABCD$.
E is a point on the side CD.
$AE = 12$ cm, angle $DAE = 30°$ and angle $EAB = 60°$.



(a) Show that the exact length of AD is $6\sqrt{3}$ cm.

[2]

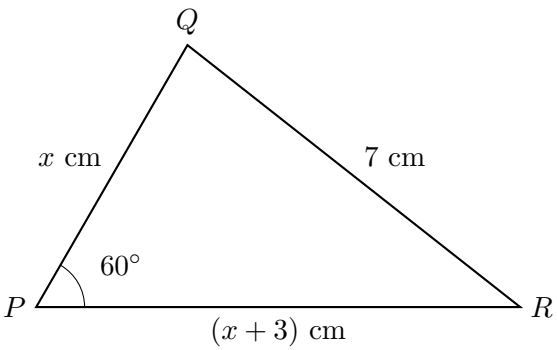
(b) The exact area of the rectangle $ABCD$ is $60\sqrt{3}$ cm^2.
Work out the exact length of CE.

..... cm **[3]**

(c) Find the exact value of $\tan(\angle EBC)$.
Give your answer in the form $\dfrac{a\sqrt{b}}{c}$ where a, b, and c are integers.

..... **[3]**

9 The diagram shows a triangle PQR.
$PQ = x$ cm, $PR = (x + 3)$ cm and $QR = 7$ cm.
Angle $QPR = 60°$.



(a) Show that $x^2 + 3x - 40 = 0$.

[4]

(b) Solve the equation to find the value of x, and hence find the perimeter of triangle PQR.

Perimeter = ... cm **[3]**

(c) Find the exact area of triangle PQR.

Area = ... cm^2 **[3]**

1 **(a)** $\frac{\sqrt{3}}{2}$ **[B1]**

(b) $\sin 30° = \frac{1}{2}$ and $\tan 60° = \sqrt{3}$ **[M1]**

$\frac{1 - 1/2}{(\sqrt{3})^2} = \frac{1/2}{3} = \frac{1}{6}$ with no errors seen **[A1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** 2 **[B1]**

(b) $\frac{1}{2}$ **[B1]**

(c) $\frac{\sqrt{3}}{\sqrt{3}/2}$ **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Substitutes $\sin 60° = \frac{\sqrt{3}}{2}$ and $\tan 30° = \frac{1}{\sqrt{3}}$ (or $\frac{\sqrt{3}}{3}$) **[M1]**

1 **[A1]**

(b) Substitutes $\tan 45° = 1$ leading to $\cos x = \frac{1}{\sqrt{2}}$ **[M1]**

45 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** $\sin 45° = \frac{6}{PR}$ (or $\cos 45° = \frac{6}{PR}$) **[M1]**

$PR = 6 \div \frac{\sqrt{2}}{2} = 6\sqrt{2}$ **[A1]**

(b) $\sin 60° = \frac{6\sqrt{2}}{PS}$ **[M1]**

$PS = 6\sqrt{2} \div \frac{\sqrt{3}}{2}$ **[M1]**

$PS = \frac{12\sqrt{2}}{\sqrt{3}} = \frac{12\sqrt{6}}{3} = 4\sqrt{6}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** $\sin 45° = \frac{PQ}{8\sqrt{2}}$ **[M1]**

$PQ = 8\sqrt{2} \times \frac{1}{\sqrt{2}} = 8$ **[A1]**

(b) $\cos 30° = \dfrac{8\sqrt{2}}{PS}$ or $PS = \dfrac{8\sqrt{2}}{\cos 30°}$ **[M1]**

$PS = \dfrac{8\sqrt{2}}{\sqrt{3}/2} = \dfrac{16\sqrt{2}}{\sqrt{3}}$ **[M1]**

Rationalises the denominator to get $\dfrac{16\sqrt{6}}{3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

6 In $\triangle ABD$, $AD = 8\sqrt{2}\cos 45° = 8$ **[M1]**

$BD = 8\sqrt{2}\sin 45° = 8$ **[A1]**

In $\triangle BDC$, $BC = \dfrac{BD}{\sin 30°} = \dfrac{8}{0.5} = 16$ **[M1]**

$DC = \dfrac{BD}{\tan 30°} = 8\sqrt{3}$ **[M1]**

Perimeter $= AB + BC + AD + DC = 8\sqrt{2} + 16 + 8 + 8\sqrt{3}$ **[M1]**

$24 + 8\sqrt{2} + 8\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

7 (a) Substitutes $\tan 30° = \dfrac{1}{\sqrt{3}}$ to get $\tan^2 30° = \dfrac{1}{3}$ **[B1]**

Evaluates numerator as $1 - \dfrac{1}{3} = \dfrac{2}{3}$ AND denominator as $1 + \dfrac{1}{3} = \dfrac{4}{3}$ **[M1]**

Simplifies $\dfrac{2/3}{4/3} = \dfrac{1}{2}$ which is equal to $\cos 60°$ **[A1]**

(b) Substitutes exact values: $x\left(\dfrac{\sqrt{3}}{2}\right) = \sqrt{3} - x\left(\dfrac{1}{2}\right)$ **[M1]**

Rearranges to collect x terms: $x\left(\dfrac{\sqrt{3}}{2} + \dfrac{1}{2}\right) = \sqrt{3}$ **[M1]**

Isolates x: $x = \dfrac{2\sqrt{3}}{\sqrt{3}+1}$ **[M1]**

Rationalises denominator: $\dfrac{2\sqrt{3}(\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)} = \dfrac{6-2\sqrt{3}}{2} = 3 - \sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

8 (a) $\cos 30° = \dfrac{AD}{12}$ **[M1]**

$AD = 12 \times \dfrac{\sqrt{3}}{2} = 6\sqrt{3}$ with no errors seen **[A1]**

(b) Uses Area $= AB \times AD = 60\sqrt{3}$ to find $AB = 10$ **[B1]**

$DE = 12\sin 30° = 6$ **[M1]**

$CE = CD - DE = 10 - 6 = 4$ **[A1]**

(c) Identifies $BC = 6\sqrt{3}$ and $CE = 4$ **[M1]**

$\tan(\angle EBC) = \dfrac{CE}{BC} = \dfrac{4}{6\sqrt{3}}$ **[M1]**

$\dfrac{2}{3\sqrt{3}} = \dfrac{2\sqrt{3}}{9}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

9 (a) Uses cosine rule: $7^2 = x^2 + (x+3)^2 - 2(x)(x+3)\cos 60°$ **[M1]**

Substitutes $\cos 60° = 0.5$: $49 = x^2 + x^2 + 6x + 9 - x(x+3)$ **[M1]**

Expands and simplifies: $49 = 2x^2 + 6x + 9 - x^2 - 3x$ **[M1]**

$x^2 + 3x - 40 = 0$ **[A1]**

(b) $(x+8)(x-5) = 0$ **[M1]**

$x = 5$ (rejects $x = -8$) **[A1]**

Perimeter $= 5 + (5+3) + 7 = 20$ **[A1]**

(c) Uses sine area formula: Area $= \dfrac{1}{2} \times 5 \times 8 \times \sin 60°$ **[M1]**

Substitutes $\sin 60° = \dfrac{\sqrt{3}}{2}$ **[M1]**

$10\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 10]*